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CLINICAL PRACTICE GUIDELINE

2025 AHA/ACC Statement on Cost/Value Methodology in Clinical Practice Guidelines (Update From 2014 Statement): A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines

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AIM: The "2025 AHA/ACC Statement on Cost/Value Methodology in Clinical Practice Guidelines (Update From 2014 Statement)" describes a systematic approach for consistent implementation of "economic value statements" across ACC/AHA guidelines. It updates the cost-effectiveness threshold and proposes a new level of certainty framework that summarizes the strength of the available evidence. Additionally, it describes how cost-effectiveness analyses (CEAs) can help advance equity in population cardiovascular health.

METHODS: A focused literature search was conducted from January 9, 2024, to February 2, 2024, encompassing English-language publications related to CEA methodology in PubMed, EMBASE, and the Cochrane Library, with publication dates ranging from 1973 to the present. Additional relevant studies published during the writing process (through June 25, 2024) were also considered by the writing committee.

STRUCTURE: This Cost/Value Methodology Statement updates prior guidance regarding the incorporation of evidence from published CEAs into clinical guidelines. It provides guidance for identifying and synthesizing relevant high-quality evidence, developing economic value statements, and communicating level of certainty in such statements. It defines the US cost-effectiveness threshold as $120 000 per quality-adjusted life year gained, highlights special considerations related to cardiovascular drugs and devices, emphasizes health equity considerations when interpreting CEAs, and defines a reference case for future CEAs.

Key Words: AHA Scientific Statements ■ cost ■ cost/value ■ cost-benefit analysis ■ cost-effectiveness analysis ■ cost-effectiveness ■ decision-making ■ economic evaluation ■ global health ■ health status ■ ICER ■ quality of life ■ quality-adjusted life years ■ QALY ■ United States ■ World Health Organization

*Writing committee members are required to recuse themselves from voting on sections to which their specific relationships with industry and other entities may apply; see Appendix 1 for detailed information. †AHA/ACC representative. ‡AHA/ACC joint staff representative.

AHA/ACC Joint Committee on Clinical Practice Guidelines Members see page e352.

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1. PREFACE

1.1. Scope

The ACC/AHA published its first cost/value methodology statement in 2014.[1] In the decade since its publication, this statement has served as a methodological guidepost for economic evaluations in cardiology and has led to the routine incorporation of "Value Statements" in ACC/AHA guidelines.

The past 10 years have seen a proliferation of publications focused on cost-effectiveness analysis (CEA) and important advances in CEA methodology, including guidance from the Second Panel on Cost-Effectiveness in Health and Medicine,[2] the introduction of novel alternatives to quality-adjusted life years (QALYs), and an increasing emphasis on patient-centered and patient-reported outcomes. Challenges in CEA methodology remain, including lack of clarity on nationally representative costs for new drugs and devices, interpretation of heterogeneity of cost effectiveness across subpopulations, and incorporation of methods to address health inequities.

This ACC/AHA methodology statement provides updated guidance for investigators, academic journals, and clinical guideline committees regarding the rigorous evaluation of CEAs. This update incorporates novel CEA methods and highlights areas where further methodological research is needed. It updates the cost-effectiveness thresholds to reflect currently available empiric evidence regarding willingness-to-pay thresholds in the United States. Importantly, the 2025 update includes a focus on health equity in CEA, with the goal of ensuring that decision-makers employ high-quality evidence in efforts to improve population health.

1.2. Structure and Membership of the Writing Committee

The members of the writing committee included experienced clinicians and specialists in cardiology, health economics, and research methodology.

1.3. Disclosure of Relationships With Industry and Other Entities

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or improper influence. The complete policy on relationships with industry and other entities (RWI) can be found online. Appendix 1 lists writing committee members' and peer reviewers' comprehensive RWI.

1.4. Abbreviations

Abbreviation	Meaning/Phrase
CEA	cost-effectiveness analysis
CHEQUE	Criteria for Health Economic Quality Evaluation
GDP	gross domestic product
ICER	incremental cost-effectiveness ratio
QALY	quality-adjusted life year

2. INTRODUCTION

2.1. Importance of CEAs

Over the past 20 years, the introduction of novel strategies for the prevention and treatment of cardiovascular disease have produced a substantial improvement in population health outcomes in the United States.[3] At the same time, uneven access and rising health care costs have led to increased scrutiny of the cost and value of interventions.[4–6] Over the next 30 years, health care costs attributable to cardiovascular disease in the United States are expected to quadruple, even after accounting for inflation—from $338 billion in 2020 to $1.4 trillion in 2050, an increase from 2.7% of the gross domestic product (GDP) in 2020 to 4.6% of the projected GDP in 2050.[7]

Because societal resources are finite, resources used to deliver 1 type of health intervention (eg, a novel treatment for heart failure) are then no longer available to provide other health care interventions (eg, screening for hypertension) or, more broadly, to address other societal priorities beyond health care. Optimal allocation of societal resources creates a need to understand trade-offs between alternative uses of available resources, because committing resources to a particular option generates the "opportunity cost" of foregone benefits that would have accrued if the resources had instead been committed to an alternative. One approach to optimizing resource allocation is maximizing efficiency (ie, achieving maximum gains in population-level outcomes per unit resource used). CEAs can help promote efficient allocation of finite societal resources by weighing the effectiveness of an intervention against the incremental use of resources.[8] It is important to note that CEAs are not simply about minimizing costs: they systematically evaluate the incremental cost-effectiveness of an intervention relative to the next best alternative to maximize the population health benefits, given finite available resources. In doing so, they produce an estimate of the efficiency of health care benefit production for that intervention to help inform decision-making at the policy and system level.

Three caveats are worth noting at the outset. First, the intention of CEAs is not to directly influence patient-level decision-making. Instead, systematically applying CEAs at the level of the health system can promote efficient use of resources and improve population health outcomes. To do so, they would ideally be performed across the entire set of options available to generate health, rather than evaluating each intervention in a silo. This can be challenging in the United States given the fragmented nature of our health care system. Alternatively, CEAs can inform the prices of new health technologies entering the market, thus improving the affordability of and access to novel diagnostic and treatment strategies.[9] Second, resource allocation is a complex process based on several considerations in addition to the efficient use of resources, including political priorities, scalability and affordability of the proposed intervention, and its impact on health equity. These considerations are particularly important for the prevention and treatment of cardiovascular disease given the large number of individuals affected—1 in 3 US adults received care for a cardiovascular risk factor or condition in 2020—and the substantial economic burden of cardiovascular disease.[7,10] Third, in order to improve health care efficiency, CEAs have to be updated over the intervention's life cycle as more cost and effectiveness data become available, and policymakers have to be willing to defund less-efficient programs already in place. Adding more programs and interventions that are "cost-effective" without removing interventions that are no longer cost-effective continues to drive up health care spending without optimizing efficiency.

This statement uses the term CEA to describe analytic frameworks with effectiveness outcomes that reflect patient or community preferences over health states (eg, QALYs). CEAs that use QALYs or outcomes measured in natural units (eg, years of life saved) comprise the majority of economic evaluations in cardiovascular medicine. Although this statement focuses on cardiovascular CEAs performed from a US health care sector or societal perspective, the underlying concepts are broadly applicable to CEAs in other settings and for other health care conditions.

3. IMPLICATIONS FOR HEALTH EQUITY

Equity is the absence of unfair, avoidable, or remediable differences among groups of people, whether those groups are defined socially, economically, demographically, geographically, or by other characteristics (eg, sex, gender, ethnicity, disability, or sexual orientation).[11] Unlike health equality, which refers to equal access to quality health care for all, health equity is achieved when everyone can attain their full potential for health

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and well-being.[12] In 2010, Marmot et al[13] noted in their review, *Fair Society, Healthy Lives*, that if society's goal is to reduce or eliminate the social gradient in health, "actions must be universal, but with a scale and intensity that is proportionate to the level of disadvantage." This principle, which they termed "proportionate universalism," has been proposed as a potential solution to reduce health disparities and advance population health equity.[14]

Health equity is a central and crucial part of health care quality, and no assessment of value is complete without consideration of equity implications. Historically, however, cardiovascular CEAs have not included quantitative evaluations of health equity, raising concerns that solely focusing on the efficient use of resources may interfere with efforts to pursue equity.[15] It is important to note that efficiency and equity are not necessarily at odds; indeed, in settings of finite resources, maximizing efficiency may make it possible to expand access to effective therapies and thus improve health equity. This is particularly true for cardiovascular disease because historically marginalized populations often bear a disproportionate burden of the disease. As a result, strategies that optimize efficiency and strategies that maximize equity often target the same high-risk underserved populations. Studies that evaluate not only the incremental cost effectiveness of a strategy across the population but that also examine the heterogeneity of cost effectiveness in subpopulations can help advance our understanding of which subpopulations should be prioritized for access and adoption. This should be guided by Roemer's Equality of Opportunity approach (ie, to "level the playing field" among US adults at risk for or living with cardiovascular disease by allocating greater resources to those at greatest risk or most likely to benefit[16]).

For evaluations that result in an efficiency-equity trade-off, methodological advances now make it possible to explicitly incorporate health equity considerations into CEAs. This can be done through the use of formal approaches that quantitatively account for the impact of an intervention on health disparities (distributional CEA) or catastrophic out-of-pocket expenditures (extended CEA).[17–19] Alternatively, health equity considerations may influence how we interpret the findings of CEAs by accepting higher willingness-to-pay thresholds for interventions that enhance equity among key populations, such as people with disabilities or chronic comorbidities, who have been historically marginalized (eg, those from racially or ethnically minoritized groups), or communities disinvested of health-promoting resources. Even where distributional or extended CEAs are not available, a holistic assessment of economic value should consider the impact of the disease on outcomes across diverse population groups.[20]

4. SELECTING HIGH-QUALITY ECONOMIC EVIDENCE FOR INCORPORATION INTO CLINICAL GUIDELINES

4.1. Which Therapies Require a Value Statement?

Consistent with the 2014 cost/value clinical practice guidelines, we suggest that clinical guidelines include economic value statements about all interventions for which economic evaluations of adequate quality are available, including preventive, diagnostic, and therapeutic strategies, and approaches to health care delivery relevant to the clinical guidelines. The decision about whether an economic value statement is appropriate should rely on the judgment of clinical and economic experts on the guideline writing committee, taking into account the clinical and population health relevance, cost, and available evidence. To the extent feasible, guideline committees should consider including economic value statements for: 1) interventions that receive Class 1 and 2A recommendations in the clinical guidelines, and 2) interventions for which an understanding of economic value is likely to influence adoption. An example of the latter would be an intervention that is expected to substantially impact total health care spending, either because it has a high cost or because a large number of individuals are likely to be eligible. When high-quality economic evaluations are scarce, economic value statements may highlight evidence gaps that can be addressed in future research (eg, "indeterminate" economic value due to a small number of studies). Thus, the first step in developing economic value statements is a review of the literature and consultation with subject matter experts in the field to identify all economic evaluations relevant to particular clinical guidelines.

4.2. Assessing the Quality of Economic Evidence

As with clinical effectiveness studies, the quality of health economic evaluations can vary substantially.[16,21,22] Therefore, each study should undergo a careful quality assessment before its incorporation into the economic value statement. Several factors determine the quality of a health economic study, including the population studied, the choice of intervention, comparator, and outcomes, and the overall methodological rigor. For instance, the generalizability of a study's findings may be limited if the study population has different demographic, clinical, and socioeconomic characteristics than the target population that is expected to adopt the intervention in the real world. Alternatively, incorrect assumptions about the underlying risk of adverse cardiovascular events, competing risk of other conditions, choice of comparators, or costs related to cardiovascular care

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(Table 1) may compromise relevance to the target population. Established quality assessment tools, such as the Criteria for Health Economic Quality Evaluation (CHEQUE),[23] can help evaluate the methodological rigor of a health economic study by providing a structured examination of key structural components of an analysis and helping define how much the study deviates from a preferred reference case (Table 2). However, expert judgment is still required, as different assessors may reach different conclusions with such tools.

4.2.1. Study Design

Both trial-based economic evaluations, which are conducted alongside a randomized controlled trial, or population-based economic evaluations, which use simulation models to transport findings from trials or epidemiological studies to a target population, can be used to inform economic value statements.[24] These approaches vary in how inputs are estimated, the representativeness of the population being modeled, and often in their analytic time horizons.[25]

Trial-based evaluations may offer the advantage of prospectively collected costs and quality-of-life data in the trial-eligible population, especially if the economic analysis was planned prospectively and the economics team had an opportunity to include key resource consumption variables on the case report form. In addition, by using empirical data regarding costs, quality of life, and outcomes, trial-based evaluations ensure that any correlations between cost and outcome are retained in the analysis. In contrast, population-based evaluations, which are typically computer simulation models that track a cohort of the treatment-eligible population (Markov cohort models) or individual members of the treatment-eligible population (microsimulation models), may more closely reflect the real-world target population and costs than a trial-based analysis. The analytic time horizon of trial-based analyses can either be limited to the follow-up of the trial or extended beyond the length of the trial using statistical or other modeling approaches. For any kind of model-based analyses, including population-based models, careful attention must be paid to key assumptions regarding cardiovascular risk in the modeled population (including change in risk with age and over time), adherence to the intervention (based on data from trials and real-world populations), effectiveness (including magnitude and duration), safety (including short-, intermediate-, and long-term adverse events), and costs (including costs of the intervention, downstream costs, and background costs unrelated to the intervention).[26]

4.2.2. Perspective

The majority of CEAs of health interventions adopt a health care sector perspective, which captures all health care–related costs and health outcomes, regardless of who pays for them. The Second Panel on Cost-Effectiveness in Health and Medicine advises that analysts also report a reference case analysis based on the societal perspective to fully capture the consequences of investing in an intervention, including outcomes outside the health care sector.[27] This approach includes time costs of seeking care, transportation costs, and costs resulting from decreased productivity in the labor market, voluntary work, and in-household production. These costs can be particularly important for cardiovascular disease CEAs because cardiovascular events can be debilitating, produce substantial productivity losses due to morbidity and premature mortality, and result in considerable caregiver burden.[7,28,29] Consideration of these effects in CEAs of cardiovascular interventions may enable a more comprehensive evaluation, especially of preventive strategies. However, the use of a societal perspective is currently uncommon in cardiovascular CEAs, largely because costs and benefits outside the health care system are often unavailable.[30,31] Economic value statements in ACC/AHA guidelines should be based on CEAs that adopted a health care sector perspective, and results from the societal perspective, if available, should be reported in the supporting text.

4.2.3. Analytic Time Horizon

Economic value statements should be informed by CEAs that adopt an analytic horizon long enough to fully capture the health effects and resource use of the intervention in question, which is usually a lifetime horizon in CEAs of cardiovascular conditions.[32] For instance, a cardiac arrest or a disabling stroke may have long-term effects on an individual's quality of life and health care spending.[33,34] Some interventions have delayed benefits (eg, the cumulative effect of a lifetime of lowering cholesterol levels on cardiovascular risk) or costs (eg, generator change of an implantable defibrillator), while others may have safety concerns that appear many years after the initial intervention (eg, endocarditis from a pacemaker lead infection).[35,36] Therefore, economic evaluations of cardiovascular interventions should generally consider the lifetime horizon as the default (to fully capture costs and benefits) with any alternatives carefully justified. However, randomized trials evaluating a novel therapy rarely follow trial subjects for more than 5 years, and the long-term effects and safety of interventions are uncertain. As a result, economic evaluations that incorporate longer analytic horizons must make assumptions about long-term effectiveness and safety based on extrapolations of available data. Therefore, interpretation of findings from such CEAs demands careful attention to the results of sensitivity analyses that examine the effect of using shorter time horizons, as well as scenario analyses that vary key assumptions regarding long-term effectiveness and safety. Similarly, although short analytic horizons may be acceptable for some questions, these studies should be carefully vetted to identify any bias resulting from short horizons. For instance, CEAs with

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short analytic horizons may make screening and prevention appear less economically attractive when costs are incurred upfront and benefits are delayed. Major cardiovascular procedures and device implants are subject to similar considerations, where upfront procedural costs and downstream health benefits are accrued at different time points in follow-up. The choice of analytic horizon should be guided by the characteristics of the study population, intervention, and disease.[32,37]

4.2.4. Study Population

Economic value statements should be informed by analyses that examine the cost-effectiveness of the intervention in the target population for which the intervention is being recommended. Often, real-world populations have a higher risk of adverse cardiovascular events than populations enrolled in clinical trials and therefore derive a larger absolute benefit from an effective intervention. At the same time, real-world populations may also have a greater burden of comorbidities, which may translate to a greater competing risk of events not affected by the intervention (eg, death from noncardiovascular causes), which may reduce the absolute benefit of the intervention compared with what was observed in the trial population. When the economic value statement must rely on studies that examine a population that is systematically different from the target population, this should be reflected in the wording of the economic value statement. For instance, if a lipid-lowering therapy is approved for use in all patients with established atherosclerotic cardiovascular disease but the relevant CEA only evaluated its cost effectiveness in patients with a recent history of myocardial infarction, the economic value statement should explicitly state that its assessment applies to the latter high-risk population.

4.2.5. Intervention and Comparator(s)

Economic value statements should be informed by CEAs that have a well-defined intervention strategy (eg, drug or device coupled with the delivery program needed for implementation) and consider all appropriate comparator strategies for the condition and population being studied. In this context, appropriateness of comparators refers to being indicated for the condition and population as well as potentially accessible to the population being evaluated. Because the cost-effectiveness of an intervention is measured relative to a comparator, the omission of a relevant comparator (or the inclusion of an inappropriate comparator) can meaningfully change the estimated incremental cost-effectiveness ratio (ICER) and produce misleading results. Judging whether a CEA incorporates the relevant comparator(s) relies on the clinical expertise of the guideline committee.

4.2.6. Costs

The underlying objective of the costing exercise is to quantify the resources required to adopt the study intervention relative to the chosen comparator. These costs have 2 components: resources used during the production and delivery of the intervention being studied, and the resources used (or saved) as a consequence of adopting the intervention.[38] High-quality CEAs incorporate a comprehensive assessment of costs, including the initial cost of the intervention, the cost of delivering the intervention and cotreatments (eg, the cost of outpatient visits necessary to escalate the dose of a therapy), subsequent treatment-related costs, including the cost of managing any adverse events, and future costs resulting from prolonged survival.[39–41] They also account for any savings from averted adverse cardiovascular events.

The range of costs that are included in the analysis depends on the perspective adopted by the study: costs outside the formal health care sector, such as caregiver time or productivity losses, are excluded from a CEA that adopts a health care sector perspective but are included in a CEA that adopts a societal perspective (Table 1). Costs should be based in the same jurisdiction as the target population.[38] Therefore, economic value statements in ACC/AHA guidelines should primarily rely on CEAs that use costs that are observed or estimated in the US population (costs measured in other populations and converted to US dollars are not usable for this purpose). Note that this prioritization of US-based estimates only applies to costs; it is common for effectiveness inputs of a cardiovascular CEA to be derived from multinational studies. The supporting statement may highlight findings from high-quality CEAs in other jurisdictions, particularly if the comparison generates important insights about the value of the intervention being examined.

Health Care Costs

Health care costs refer to the resources consumed in the production of health care that are then no longer available for other uses. These include costs of drugs, devices, durable medical equipment, clinician and nonclinician time, inpatient hospitalization, emergency room visits, outpatient services, long-term care, home-based services, etc. Broadly, there are 2 approaches to evaluating health care costs[38]:

1) Micro-costing, which is a bottom-up approach that relies on direct enumeration and costing of all resources consumed or saved
2) Gross-costing, which is a top-down approach that assigns each episode of care an average aggregate cost based on known national reimbursement rates (eg, Medicare reimbursement for a Diagnosis-Related Group)

The choice of approach used to estimate costs depends on the specific types of resources to be valued.[38] Micro-costing is necessary when available gross-costing estimates do not fully capture the difference in the resource utilization between the 2 arms.[42] For instance, when evaluating a new (ie, currently

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Table 1. Cost Components Included in a Cost-Effectiveness Analysis Based on the Perspective

Cost Component	Examples	Perspective		
		Health Care Sector	"Modified Societal"*	Full Societal
Health care costs paid by third-party payers	Drug/device costs, hospital services, physician services, home care services, allied health, etc.	X	X	X
Costs paid out-of-pocket by patients	Drug copayment/coinsurance, user fees for inpatient or outpatient services, etc.	X	X	X
Patient-time costs	Time spent seeking care		X	X
Unpaid caregiver-time costs	Time spent in caregiving		X	X
Transportation costs			X	X
Productivity	Formal/informal labor earnings lost Cost of unpaid lost productivity due to illness Cost of uncompensated household production		X	X
Social services				X
Housing				X
Legal or criminal justice				X
Education				X
Environmental				X

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*Analysts should attempt to capture all relevant costs from a societal perspective. When some relevant aspects of societal perspective cannot be measured or quantified, a "modified" societal perspective may be used. This modified societal perspective includes some societal costs, such as patient time and lost productivity, but excludes costs from other sectors of the economy. An Impact Inventory can aid in communicating and discussing these limitations.

unreimbursed) device such as a drug-coated balloon for use during a percutaneous coronary intervention, the average Medicare reimbursement for a percutaneous coronary intervention would obscure the difference in resources between the control and intervention arms, and micro-costing is necessary to capture differences in device costs, catheterization laboratory time, operator time, contrast use, etc. In other contexts, national average cost estimates may be sufficient and preferred over micro-costing estimates given the imprecision of cost estimates derived from finite samples.

When the gross-costing approach is used, estimates should reflect actual amounts paid, including out-of-pocket costs. This can be challenging in settings like the United States, where there is wide variation in reimbursement rates among the different private and public payers. The age of the population affected should be considered when sourcing cost inputs: use of Medicare reimbursement rates is reasonable for individuals who are 65 years or older, whereas commercial and Medicaid reimbursement rates are preferred for individuals younger than 65 years. Because list prices ("charges") for hospital services are greater than the opportunity cost of the resources used to provide the service and do not reflect the actual amounts paid, they should be adjusted by using appropriate cost-to-charge ratios to estimate costs. When using individual-level observed costs (eg, costs accrued by each patient over the course of follow-up), statistical analyses should account for the fact that costs are positively skewed and heteroskedastic.[38]

When costs related to the condition being studied are easily identifiable and the intervention of interest does not affect survival, it is acceptable to only focus on costs related to the condition for the estimation of incremental health care costs. However, when costs cannot easily be separated into being related or unrelated to the condition being treated, or when survival under the intervention and comparator strategies differ, the preferred approach is to consider all health care costs, including those unrelated to the condition being evaluated.[38,39]

Health care cost inputs from older studies should be inflated using the Personal Health Care Expenditure deflator developed by the Centers for Medicare and Medicaid Services or the Personal Consumption Expenditure price index developed by the US Bureau for Economic Analysis.[43,44] Note that the medical care component of the Consumer Price Index excludes expenditures by third-party payers, which makes it a biased indicator of overall inflation in health care costs, and is unsuitable for use in CEAs.[45]

Intervention Costs

Recognizing the complexity of drug and medical device pricing in the United States, high-quality economic evaluations should incorporate an estimate of net price to the payer after discounts or rebates in their base-case analysis. Ideally, this reflects the net price to the payer or payers for the entire or majority of the treatment-eligible population being evaluated, rather than a specific subpopulation.[46] Because the magnitude of discounts and rebates can vary substantially by payer and because

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prices may vary by health system and over time, CEAs should perform sensitivity analyses that vary the cost of the intervention over a range that includes higher and lower discounts compared with the base case, the list price at the time the study was conducted, and future price increases. The results of such a sensitivity analysis may facilitate the identification of threshold costs at which an intervention would meet commonly used cost-effectiveness thresholds; these may be reported in the economic value statement or the supporting text. Any costs related to the administration of the treatment (eg, intravenous supplies, nursing time, device implantation costs) should also be included in the analysis. Where appropriate, treatment costs should also be adjusted for nonadherence over time. See Section 8, "Key Challenges and Special Considerations," for additional details regarding estimating the cost of medical devices and rare disease therapies.

Future Costs

The Second Panel on Cost-Effectiveness in Health and Medicine advises that CEAs conducted from a health care sector perspective should also capture all future costs, that is, total health care sector costs during the added years of life, including health care costs unrelated to the disease of interest (see Basu[38] for a detailed discussion of the theoretical underpinnings about future costs). Additionally, CEAs from a societal perspective should also include non–health care consumption costs during the added years of life.

Other Costs

In addition to these health care costs, analyses that adopt a societal perspective should include illness- or treatment-related productivity losses resulting from decreased participation in formal and informal labor markets or change in household production, the value of the patient's time spent seeking care as well as any caregiver time, and, where relevant, any costs borne by other sectors of the economy, such as education or criminal justice (Table 1). Sometimes not all aspects of a societal perspective can be measured or quantified, in which case an impact inventory can be used to discuss the limitations. For instance, a modified societal perspective may include some societal costs, such as patient time and lost productivity, but exclude costs, such as education, from other sectors of the economy.[47]

4.2.7. Health Outcomes

The health outcomes of an intervention relative to the comparator may be a change in the incidence of clinical events (eg, number of myocardial infarctions averted with lipid-lowering therapy) or change in survival (ie, life years gained). The use of the intervention itself may affect health-related quality of life (eg, the disutility associated with taking a daily pill or a weekly injection), as may minor or major adverse events related to the intervention (eg, myalgia or injection site reactions).

Ideally, the chosen outcome fully captures the effect of an intervention on survival as well as that on health-related quality of life. The QALY, a commonly used measure of health benefits in economic evaluations, is calculated by multiplying the health-related quality of life associated with a health state with the time spent in that state.[48] Typically, health-related quality of life is measured on a scale from 0 to 1, where 0 represents death and 1 represents perfect health. The QALY provides a common metric in which different health outcomes can be measured and compared (eg, a myocardial infarction and a gastrointestinal bleed), enabling comprehensive evaluation of all the outcomes related to an intervention or disease state, as well as decision-making across different programs. Health-related quality of life weights elicited from a representative sample of fully informed members of the community are generally preferred over those elicited from patients experiencing chronic illness because patients adapt to their condition and typically report higher estimates of health-related quality of life than preferences elicited from the community.[48–50] However, some experts have argued that economic and ethical principles support the use of quality-of-life estimates elicited from patients who are directly affected by the condition.[51]

There are several important limitations to using the QALY as an outcome in CEAs, most notably that gain in survival among individuals with a higher baseline quality of life generates more QALYs than an equivalent gain in survival among individuals with a lower baseline quality of life.[52] This translates to a comparatively higher valuation of interventions that prolong survival in populations with a better baseline health status, raising concerns about the potential of QALY-based resource allocation to exacerbate health disparities. Another key limitation is the assumption that subjects are indifferent to the choice of a change in quality of life compared with a change in longevity that yields the same net change in QALYs, which is often not the case.

Several alternative metrics have been proposed to address these concerns, including the "equal value of life-years gained"[53] and the "health years in total."[54,55] Although a complete discussion of the strengths and limitations of these new approaches is beyond the scope of this statement,[56,57] it is important to note that these metrics have not been uniformly adopted by major health technology assessment agencies, cost-effectiveness thresholds have not been established for some of these newer metrics, and the results of studies that use these outcomes are not directly comparable to the results of studies that report incremental cost per QALYs gained.

Economic value statements should incorporate findings from available CEAs that measure health outcomes in terms of QALYs and life years, and, where available,

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equal value of life years or health years in total. While we recognize that the QALY is an imperfect measure, it provides useful insights about the efficiency of resource allocation. Moreover, in practice, adjusting life years for quality of life does not substantially alter the results of CEAs for many interventions, including among older adults.[58,59] Any resulting concerns about health equity may be addressed using the approaches described in Section 3, "Implications for Health Equity." If alternative metrics and the corresponding cost-effectiveness thresholds are more widely adopted in the future, it would be reasonable to incorporate them into economic value statements or supporting text.

4.2.8. Effectiveness and Safety

Economic value statements should be based on CEAs in which the effectiveness of the intervention relative to the comparator is estimated from either high-quality randomized trials or observational studies that use a study design amenable to causal interpretation. Because the ICER is typically sensitive to the effect of the intervention on mortality, careful attention should be paid to assumptions regarding the point-estimate and uncertainty interval of the effect of the intervention on mortality. Another consideration is the adherence to the intervention in the study from which the effectiveness estimates are drawn. When rates of nonadherence in the source studies are high, the intention-to-treat estimates may underestimate the effectiveness of the therapy in the treated, thus making the intervention appear less economically attractive than it truly is.[60] CEAs should use effectiveness inputs from studies with high rates of adherence to the assigned intervention and control strategies; when such studies are unavailable, analysts should consider adjusting the intention-to-treat effectiveness estimates for rates of nonadherence observed in the study to estimate the effectiveness of the intervention among those who receive the intervention (ie, the local average treatment effect).[61–63] For interventions that require sustained adherence, the CEA should incorporate realistic rates of treatment discontinuation over time, either in the base case or in sensitivity analyses. High-quality CEAs should capture the effect of key treatment-related adverse events on costs and health outcomes, varying the prevalence of adverse events in sensitivity analyses.

4.2.9. Discounting

The economic value statement in ACC/AHA guidelines should use studies that incorporate discounting of future costs and health effects, so that the ICER is computed based on the present value of costs and health effects. The theoretical basis for discounting in CEAs is presented elsewhere,[64] but the rationale is that decision-makers want to maximize health (or utility) while assessing the present value—rather than the future value—of costs and effectiveness. Discounting enables appropriate adjustment for the differential timing of costs and effects. The current guidance is for CEAs adopting US health care sector or societal perspectives to apply a 3% per year discounting rate for future costs and health outcomes and vary the discount rate from 2% to 9% in sensitivity analyses.[64] If varying the discount rate produced a meaningful difference in the findings of a CEA, this may be reported in the supporting text of the guideline.

4.2.10. Cost-Effectiveness Metrics

The most widely used metric of cost-effectiveness is the incremental cost-effectiveness ratio or ICER, defined as the ratio of the net increase in spending of the intervention versus the comparator to the net increase or improvement in outcome.

In a pairwise comparison, interventions that produce higher costs and better health outcomes should be reported as cost-effective if the ICER is less than the cost-effectiveness threshold or not cost-effective if the ICER is greater than or equal to the cost-effectiveness threshold (Figure 1). Economic value statements should include both the point estimate as well as the probability of being cost-effective at the accepted cost-effectiveness threshold when available. Although it is reasonable for the supporting text accompanying the economic value statement to report uncertainty intervals around the ICER when all results are within 1 quadrant of the cost-effectiveness plane, the probabilistic findings (ie, the probability that the ICER is at or below the cost-effectiveness threshold) may be easier to interpret. Incremental CEA methods should be used when evaluating 3 or more mutually exclusive interventions.[65]

Of note, interventions that are less effective and have lower costs relative to the comparator (ie, interventions that are in the lower-cost/lower-effectiveness quadrant of the cost-effectiveness plane) are considered cost-effective if the ICER is *greater than* the cost-effectiveness threshold. This situation does not arise often, because patients, clinicians, and health systems rarely accept a new intervention that produces worse health outcomes compared with the current standard of care, even if it reduces spending. But when this situation does arise, it necessitates careful interpretation of the ICER (both the estimate and the uncertainty interval) to determine whether the intervention is cost-effective.[66,67]

Interventions that produce lower costs and better health outcomes than the comparator should be reported in economic value statements as cost-saving (or "dominant"); interventions that produce higher costs and worse health outcomes than the comparator should be reported in economic value statements as inferior (or "dominated") (Figure 1). Negative ICERs should not be reported because they are not interpretable.

As seen in Figure 1, interpretation of the ICER requires contextual information about its location on the cost-effectiveness plane relative to the cost-effectiveness threshold. This complicates statistical

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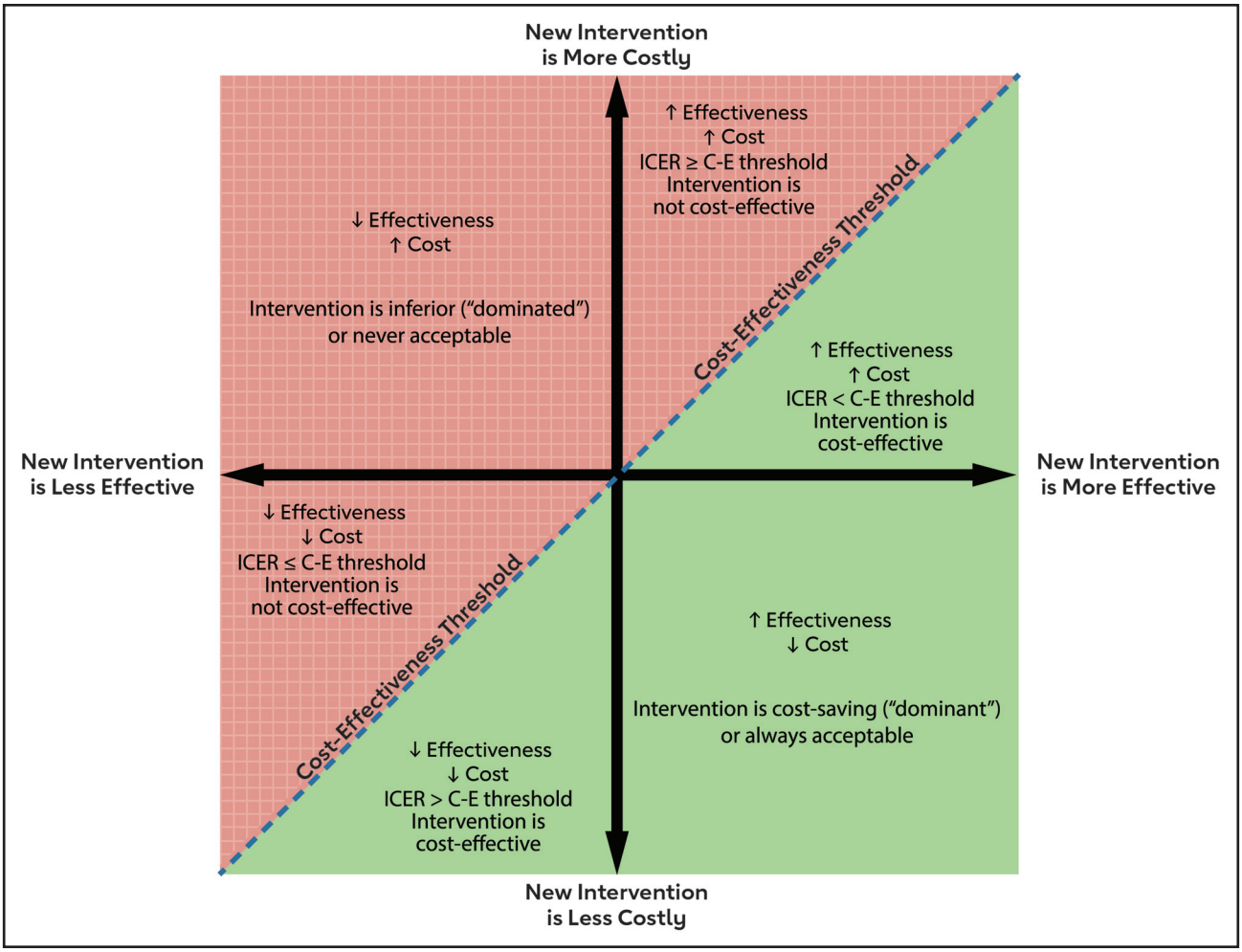

Figure 1. The Cost-Effectiveness Plane.
When a new intervention produces less health and lower costs compared with the comparator (ie, is in the lower-cost/lower-effectiveness quadrant of the cost-effectiveness plane), it may be considered cost-effective if the ICER is greater than the cost-effectiveness threshold. This situation does not arise often, because patients, clinicians, and health systems rarely accept a new intervention that produces worse health outcomes compared with the current standard of care, even if it reduces spending. Adapted with permission from Stinnett et al.[66] Copyright 1998 Sage Publications and Society for Medical Decision Making. C-E indicates cost-effectiveness; and ICER, incremental cost-effectiveness ratio.

inference when interpreting uncertainty intervals that spill into several different quadrants. CEAs typically separately report uncertainty intervals for costs and outcomes, which can provide additional insights into the uncertainty of the core components of the ICER estimate. Net monetary benefit and net health benefit are alternative metrics of cost-effectiveness that have more favorable statistical properties (Figure 2).[66] Key advantages of these metrics over the ICER include that they are calculable for all interventions, including cost-saving and

$$ICER = \frac{\text{Cost of Intervention} - \text{Cost of Comparator}}{\text{Health Outcomes with Intervention} - \text{Health Outcomes with Comparator}}$$

$$NMB = (\text{Incremental Health Outcomes} * \text{CE threshold}) - \text{Incremental Costs}$$

$$NHB = \text{Incremental Health Outcomes} - \frac{\text{Incremental Costs}}{\text{CE threshold}}$$

Figure 2. Commonly Used Metrics in CEAs.
CE indicates cost-effectiveness; ICER, incremental cost-effectiveness ratio; NHB, net health benefit; and NMB, net monetary benefit.

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dominated interventions, and statistical inference is more straightforward than in the case of the ICER. Both net health benefit and net monetary benefit should be evaluated over a range of cost-effectiveness thresholds and reported as a function of the cost-effectiveness threshold (eg, NHB_{120000} is the net health benefit at a threshold of $120 000 per QALY gained).[66] The relevant decision rule is that any intervention with a net monetary benefit >0 or net health benefit >0 at a particular cost-effectiveness threshold would be considered cost-effective at that threshold. The optimal strategy when evaluating 2 or more mutually exclusive interventions is the option with the highest net monetary or net health benefit.

4.2.11. Budget Impact

Budget impact refers to the effect of adopting the intervention on total health care expenditures, accounting for both increased spending due to intervention-related costs and downstream savings from improved health in the treated population. In order to illustrate the drivers of budget impact, high-quality economic evaluations typically also separately report changes in intervention-related costs, the cost of cardiovascular care (eg, savings from averted major adverse cardiovascular events), and the cost of noncardiovascular care (eg, increased spending due to prolonged survival). The budget impact is directly proportional to the number of eligible patients who receive the intervention, with studies typically reporting the projected effect on health care spending if all eligible patients were to receive the drug. Because the budget impact may inform strategies for adoption of the intervention at a population level, this information may be included in the supporting text accompanying the economic value statement where available.

4.3. Reference Case

With the goal of promoting quality and enhancing comparability across CEAs, the Second Panel on Cost-Effectiveness in Health and Medicine suggested a set of methodological considerations to define a reference case for use by CEA investigators.[27] We have adapted this reference case for cardiovascular CEAs to facilitate comparability across cardiovascular CEAs and incorporation of high-quality cost-effectiveness evidence into clinical guidelines (Table 2).

5. INTERPRETING AVAILABLE EVIDENCE

5.1. Cost-Effectiveness Thresholds

The 2014 ACC/AHA cost-value methodology statement provided thresholds to identify high-, intermediate-, and low-value care that were based on the per capita GDP of the United States.[1] This approach of tying cost-effectiveness thresholds to per capita GDP was initially promoted by the Commission on Macroeconomics and Health and subsequently popularized by the World Health Organization's Choosing Interventions that are Cost-Effective program, which suggested an intervention that costs less than 3 times the national annual GDP per capita per disability-adjusted life year averted should be considered cost-effective, whereas an intervention that costs less than 1 times the national annual GDP per capita per disability-adjusted life year averted should be considered highly cost-effective.[69–71] Based on the per capita GDP of the United States in 2012, the writing committee of the 2014 statement determined that an intervention with an ICER <$50 000 per QALY gained should be considered high value, $50 000 to <$150 000 (1-3× per capita GDP) per QALY gained should be considered intermediate value, and ≥$150 000 per QALY gained should be considered low value.[1] However, experts in health economic evaluation have since criticized the approach of anchoring cost-effectiveness thresholds to per capita GDP, arguing that the use of these thresholds lacks theoretical justification.[72] They note that although willingness to pay for health care increases with income, there is little evidence that the relationship is linear, and other factors such as affordability and competing societal priorities are also relevant. Moreover, GDP-based benchmarks for CEA, "set such a low bar for cost effectiveness that very few interventions with evidence of efficacy can be ruled out."[72] In response, World Health Organization experts now also advise against the use of per capita GDP-based cost effectiveness thresholds and instead suggest considering cost-effectiveness information alongside other contextual information in a transparent multicomponent decision process.[73,74] The writing committee reviewed alternative approaches to incorporating CEA results into guidelines, including relying on US-based studies for identifying a cost-effectiveness threshold or thresholds for application to cardiovascular CEAs.

Vanness et al[75] used the health opportunity cost approach to estimate a cost-effectiveness threshold for the United States. The underlying intuition is as follows. When health care expenditures increase, the costs are passed back on to the population as increases in insurance premiums. This increase in premiums causes some individuals to forego purchasing health insurance, the loss of which results in increased short-term mortality and morbidity. This chain of outcomes can be used to estimate how much individuals value an additional QALY. Vanness et al[75] estimated that at 2019 prices, for every $10 000 000 increase in health care expenditures, 1860 persons were projected to become uninsured, resulting in 81 QALYs lost due to mortality and 15 QALYs lost due to morbidity, implying a cost-effectiveness threshold of $104 000 per QALY ($51 000 to $209 000 per QALY) in 2019 US dollars. In the probabilistic analysis, 40% of the simulations suggested the threshold was less than $100 000 per QALY, and 14% suggested that

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Table 2. Reference Case for High-Quality CEAs

Study design	Both trial-based economic evaluations or population-based economic evaluations can be used to inform economic value statements.
Perspective	The reference case should adopt a health care sector perspective; results from the societal perspective or a modified societal perspective should be performed if relevant data are available.
Analytic horizon	The analytic horizon should be long enough to fully capture meaningful differences in health outcomes and costs between the intervention and comparator arms. The reference case should consider adopting a lifetime analytic horizon; any exceptions should be accompanied by a clear rationale. Because of the uncertainty of extrapolating short-term evidence to long-term horizons, studies should report sensitivity analyses that examine the effect of varying the time horizon and key assumptions on which the long-term extrapolation is based.
Population	The reference case study population should be as similar as possible to the target population for which the intervention is being recommended. Subgroup analyses that explore variation in cost effectiveness in key subpopulations should also be reported, especially if the findings are meaningfully different from those seen in the entire population.
Intervention	The intervention should be clearly defined, including details such as mode of implementation, any essential concurrent diagnostics or therapies, and a follow-up protocol if relevant.
Comparator	The reference case should consider all appropriate comparator(s) (ie, those that are indicated for the condition and population being evaluated and potentially accessible to the population being evaluated). The choice of comparator(s) is based on expert clinical judgment. It is not appropriate to exclusively compare the intervention with a comparator that is known to have high costs or low effectiveness ("comparator gaming"), as this would artificially lower the ICER of the intervention (ie, make it appear more economically attractive).
Cost	Costs should be based in the same jurisdiction as the target population. Health care cost estimates should reflect actual amounts paid, including out-of-pocket costs. Micro-costing (including the use of time-driven activity-based costing as appropriate) can produce reliable cost estimates but is not always feasible. When micro-costing is not available or feasible, gross-costing is an acceptable alternative. Cost of drugs should reflect price after discounts or rebates in the base-case analysis, preferably the price available to the entirety or majority of the treatment-eligible population. Costs of medical devices should be the price net of discounts where available or the list prices when the net price is not available, and a wide range of costs should be examined in sensitivity analyses. Cost estimates should be adjusted for nonadherence, particularly if relying on intention-to-treat effectiveness estimates. The currency and year of evaluation should be clearly identified.
Health outcomes	The reference case should report outcomes in terms of life years and QALYs gained. Alternative metrics such as equal value of life year gained and health years in total should be reported when available. Intermediate outcomes (eg, number of clinical events) should be reported. Disutility related to taking the treatment (eg, pill disutility) may be included in sensitivity analyses.
Effectiveness and safety	Effectiveness and safety inputs may be drawn from high-quality randomized trials or high-quality observational studies (the latter should use study designs that support causal inference). Expert judgment should be used when identifying the base-case value of key inputs (eg, effect on cardiovascular mortality), as these should reflect the current understanding of the intervention's effectiveness. For instance, studies powered to detect a change in a composite endpoint may be underpowered to detect an effect on mortality. When the effect on mortality is not statistically significant, it may be acceptable to rely on the point estimate if this is concordant with the overall effect size (ie, assuming the same effect as on other cardiovascular endpoints) or to assume no effect on mortality, depending on the context. Effect sizes should be adjusted for adherence if there is substantial cross-over between the 2 groups, as intention-to-treat estimates may be biased to the null in this context. All important treatment-related adverse events should be included in the analysis. Key effectiveness parameters should be subjected to extensive deterministic and probabilistic sensitivity analyses.
Discounting	Future costs and benefits should be discounted at a rate of 3% per year, with a range of 2%-9% examined in sensitivity analyses.
Model validation	Event rates in the comparator arm should be externally validated against event rates observed in real-world data, epidemiological cohorts, or publicly available trial data. Event rates in the intervention arm should be validated against appropriate sources, which may include internal validation.
Results	The reference case should report the point estimate of the ICER as well as the probability of being cost-effective at the accepted cost-effectiveness threshold. It is acceptable to report uncertainty intervals in the supporting text. Negative ICERs should not be reported; instead, interventions should be reported as inferior/dominated or cost-saving/dominant as appropriate. The reference case may report the net monetary benefit, net health benefit, or both, as a function of the cost-effectiveness threshold (eg, NHB_{120000}). The use of net monetary benefit and net health benefit metrics is encouraged as they are calculable for all interventions (including cost-saving and dominated interventions), and their statistical properties facilitate estimation of uncertainty. The reference case should report intermediate outcomes (eg, number of major adverse cardiovascular events averted) and disaggregated results (eg, disaggregating health care costs into pharmaceutical costs, costs of cardiovascular care, costs of noncardiovascular care).
Sensitivity analyses	Deterministic analyses (1- and 2-way) can help quantify the impact of uncertainty in individual parameters on cost-effectiveness. Results are typically presented as graphs (eg, Figure 4) or tornado diagrams. Probabilistic sensitivity analyses typically vary a large number of input parameters simultaneously, with each parameter drawn (with replacement) from a prespecified statistical distribution that reflects the uncertainty in that parameter. Results are typically presented as proportion of simulations that meet a cost-effectiveness threshold or acceptability curves.
Reporting	The Second Panel's standardized reporting checklist should be followed while reporting the results of CEAs.[2,68]

CEA indicates cost-effectiveness analysis; ICER, incremental cost-effectiveness ratio; NHB, net health benefit; and QALY, quality-adjusted life year.

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the threshold exceeded $200 000 per QALY.[75] Updating the analysis to 2024 US dollars using the Personal Consumption Expenditure price index (health component) suggests a cost-effectiveness threshold of $117 000 per QALY (Figure 3).

Phelps[76] used a welfare economics approach (a very distinct approach from that used by Vanness et al[75]) to estimate the cost-effectiveness threshold for the United States. Their key assumption was that utility can be specified as a function of income, which they calibrated using estimates of relative risk aversion. Assuming that individuals with typical aversion to financial risk balance expenditures on health and other consumption over time to maximize their expected well-being, individuals with a per capita disposable income of $60 300—as was the case for the United States in March 2024—would be expected to be willing to pay approximately $142 000 (range: $120 000-$175 000) for 1 QALY.

Pichon-Riviere et al[77] estimated cost-effectiveness thresholds for 174 countries based on a novel conceptual framework to assess how the adoption and coverage of new interventions with a given ICER will affect the rate of increase of health expenditures per capita and life expectancy at the population level. Using 2019 World Bank data, their approach estimated a US cost-effectiveness threshold of $95 958 ($81 672-$120 181) per QALY. Corrected for the growth in the US GDP since 2019, this translates to a threshold of $112 000 ($96 000-$141 000) per QALY. Of note, these studies estimated cost-effectiveness thresholds using demand-side approaches, which typically reflect a societal perspective.

While no single method of estimating a cost-effectiveness threshold can be considered the "gold standard," the fact that these 3 disparate approaches produced similar results is encouraging.

Given the available evidence, the consensus of the writing committee is to recommend the use of $120 000 per QALY as the cost-effectiveness threshold for clinical guidelines. This threshold should be periodically reexamined, but to ensure consistency across published guidelines, this threshold should not be adjusted annually for inflation. The consensus of the writing committee is that the threshold should be revised every 10 years (or sooner in case of any large shocks to the health care economy of the United States that cause a substantial shift in health care costs).

5.2. Level of Evidence

Results from CEAs are only as reliable as the studies that produced them. In some ways, assessing the quality of evidence from CEAs is similar to evaluating the quality of evidence from comparative effectiveness studies.[78,79] For example, CEA models are informed by randomized controlled trials or observational studies, and the quality within and across these study types can vary greatly.[80] As noted above, the quality of CEAs can be assessed using a standardized approach, such as the 24-item CHEQUE tool, which assigns points based on whether certain components were included (such as validation, probabilistic sensitivity analyses, and so on), giving more weight to the elements deemed to be more important (eg, model validation receives 8 points, while probabilistic sensitivity analysis receives 3 points in the CHEQUE tool scoring system).[23] Beyond using standardized assessment instruments, CEAs may also be evaluated based on other best practice criteria. For example, simply performing model validation satisfies the criterion on the CHEQUE tool or Consolidated Health Economic Evaluation Reporting Standards checklist,[81] but methodology experts should examine the validation process (eg, the sample size and relevance of the validation data, corrections for optimism bias, and so on). Similarly, expert judgments should also be used to assess other components of CEA quality, such as study design,[82] model input parameterization, evaluation of heterogeneity,[83] and appropriateness of extrapolation techniques where relevant. Studies that assess cost effectiveness based on the output of a black box model with no demonstration of model calibration, validation, and intermediate outputs should be viewed with caution.

It is important to pay attention to the source of funding of a study, including conflicts of interest among the authors, but this should not be considered a sole criterion of study quality. Researchers have previously found that industry-funded CEAs make more favorable assumptions than CEAs funded from other sources and report favorable ICERs for higher-cost strategies even though industry-funded CEAs have similar study quality ratings compared with other studies.[84–86] A related consideration is the degree of independence of the authors from the manufacturer of the device or drug being evaluated (ie, whether the study was performed by investigators independent from the sponsor, academic investigators under institutional contract from the sponsor, a contract research organization hired by and directed by the sponsor, or analysts directly employed by the sponsor). Investigators with close commercial ties with the sponsor may have difficulty reporting results that are unfavorable to the manufacturer.

Ideally, economic value statements should rely on high-quality studies (see Section 4.2, "Assessing the Quality of Economic Evidence" and Table 2). When multiple CEAs are available on a given topic, results of high-quality CEAs should be qualitatively synthesized by a guideline writing committee member with methodology expertise in health economic evaluations. Where relevant, the accompanying supporting text should clarify the underlying rationale for why some studies were considered more relevant than others.

There is disagreement among experts about whether results across different CEAs can be quantitatively

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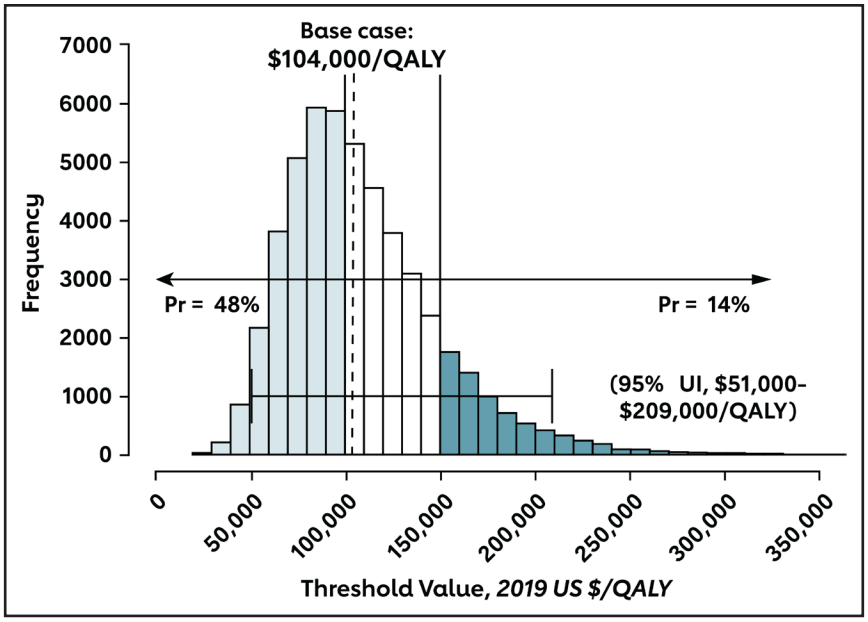

Figure 3. Using the Health Opportunity Cost Approach to Estimate the Cost-Effectiveness Threshold for US Health Care. Vanness et al[75] used the health opportunity cost approach to estimate a cost-effectiveness threshold for the United States. The figure shows the frequency of the computed cost-effectiveness thresholds in 50 000 simulations with varying input values. The light green shaded area contains 23 902/50 000 (48%) threshold values <$100 000/QALY, and the dark green shaded area contains 7006/50 000 (14%) threshold values >$150 000/QALY. The horizontal error bar shows the 95% UI. The vertical dashed line depicts the base-case estimate of $104 000/QALY in 2019 US dollars. Updating the analysis to 2024 US dollars using the Personal Consumption Expenditure price index (health component) suggests a cost-effectiveness threshold of $117 000 per QALY. Pr indicates probability; QALY, quality-adjusted life year; UI, uncertainty interval, and US, United States. Reprinted with permission from *Annals of Internal Medicine*, Vanness et al.[75] Copyright 2021 American College of Physicians. All Rights Reserved.

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synthesized using techniques such as meta-analysis; when doing so, careful attention should be paid to similarities and differences among the studies with regard to the study question, population, intervention, comparator, outcomes, model structure, and input parameters. After careful consideration, if the decision is made to proceed with a meta-analysis, costs and outcomes should be meta-analyzed separately (ie, ICERs should not be directly combined across studies). However, most guideline writing committees are unlikely to have the resources for a de novo meta-analysis of available CEAs for each question of interest. So, a comprehensive qualitative assessment of the totality of CEA evidence may be reasonable prior to forming judgments on the economic value of a given intervention or strategy. Ideally, differences in conclusions across CEAs for the same intervention should be explained by differences in modeling approach, input values, target population, and other analytic assumptions; experts will then need to judge which analytic choices were most appropriate and base their conclusions on these overall assessments.[87]

6. INTEGRATING AVAILABLE EVIDENCE INTO CLINICAL GUIDELINES

In order to clarify that cost-effectiveness is only 1 component of value (alongside total clinical benefit, feasibility, and ethical/equity considerations), we propose the use of the term "Economic Value Statement" in the guidelines instead of the term "Value Statement."

6.1. The Structure of Economic Value Statements in Clinical Guidelines

In the interest of consistency within a set of clinical guidelines and comparability across different guidelines, we suggest that economic value statements report 3 components: 1) Class of Recommendation; 2) Level of Certainty; and 3) Recommendation.

Because the economic value statement is meant to be interpreted in the context of the clinical recommendation, the economic value statement should ideally be placed immediately following the clinical recommendation for the intervention of interest.

6.1.1. Class of Recommendation

In line with the updated cost-effectiveness threshold proposed in this statement, we propose an updated Class of Recommendation framework for economic value statements in clinical guidelines (Table 3).

6.1.2. Level of Certainty

ACC/AHA clinical guidelines report level of evidence based on the number and type of studies on which the recommendation is based.[78] The framework used for clinical recommendations does not appear to fully capture the strength of evidence for economic value statements, and

Table 3. Class of Recommendation Framework for Economic Value Statements in ACC/AHA Guidelines

Economic Value Statement	Definition
Cost-saving	Intervention costs less than the comparator and produces better health outcomes.
Cost-effective	Intervention costs more than the comparator and produces better health outcomes; the ICER is less than the cost-effectiveness threshold of $120 000 per QALY gained.*
Not cost-effective	Intervention costs more than the comparator and produces better health outcomes; the ICER is greater than or equal to the cost-effectiveness threshold of $120 000 per QALY gained.*
Inferior	Intervention costs more than the comparator and produces worse health outcomes.
Indeterminate	Economic value examined but data are insufficient to draw a conclusion due to lack of studies, low-quality studies, conflicting studies, or prior studies that are no longer relevant.
Not assessed	Economic value not assessed by writing committee.

*Interpretation of the ICER is reversed when the intervention produces lower costs and less health than the comparator, see Section 4.2.10 and Figure 1 for details.

ICER indicates incremental cost-effectiveness ratio; and QALY, quality-adjusted life year.

most economic value statements in recent guidelines have been assigned a level of evidence B-NR (moderate-quality evidence from nonrandomized studies). We therefore propose an alternative framework for level of certainty as it relates to economic value statements (Table 4). This framework captures the nuances of the number and quality of CEAs available as well as how sensitive results are to uncertainty in key input parameters (eg, the effect of the intervention on mortality). The rationale for the stated level of certainty should be provided in the supporting text. Note that the level of certainty should not include considerations of whether the cost effectiveness of the intervention is likely to change if the cost of the intervention itself changes over time, which may be reported in the recommendation or the supporting text.

For interventions where the economic value is deemed to be indeterminate, no level of certainty is reported; instead, the parenthetical clause should contain information about why the class of recommendation was indeterminate (eg, insufficient evidence [lack of studies or presence of low-quality studies], conflicting evidence, or outdated evidence [prior studies that are no longer relevant]).

6.1.3. Recommendation

To the extent possible, the Recommendation should adhere to 1 of 3 formats shown in Table 5.

Format 1 reports the ICER of the intervention relative to the comparator at a given cost of the intervention, which should be used when the cost of the intervention is known. Format 2 reports the cost at which the intervention would meet the cost-effectiveness threshold, which may be particularly useful when the cost of the intervention is uncertain or variable. Format 3 combines elements of both Format 1 and Format 2. Mock-ups of economic value statements are shown in Table 6.

Table 4. Level of Certainty Framework for Economic Value Statements in ACC/AHA Guidelines*

Level of Certainty	Definition
High level of certainty	Conclusions based on findings from 1 or more high-quality CEAs with extensive sensitivity analyses suggesting that findings are robust to uncertainty in key input parameters.†
Moderate level of certainty	Evidence is deemed to be of moderate quality as a result of factors such as the number or quality of available CEAs, uncertainty of underlying evidence (eg, small size of effectiveness study), or findings that are sensitive to uncertainty in key input parameters.

*For interventions where the economic value is deemed to be indeterminate, no level of certainty is reported. Instead, the parenthetical clause should contain information about why the class of recommendation was indeterminate (eg, insufficient evidence [lack of studies or presence of low-quality studies], conflicting evidence [inconsistency of findings across individual high-quality studies that cannot be explained by methodological differences], or outdated evidence [prior studies that are no longer relevant]).

†Not including future changes in price of the comparator.

CEA indicates cost-effectiveness analysis.

6.2. Supporting Text

Supporting text should include the following key elements while remaining within guideline length requirements:

a) A summary of the high-quality CEAs that were used to generate the economic value statement. When multiple CEAs are available on a given topic, the supporting text should clarify the underlying rationale for which studies were considered to be of sufficient quality to be incorporated into the guidelines (and provide the underlying rationale for choosing these studies). These summaries should include the study population (and whether it differs from the corresponding population for the guideline statement), the intervention, the comparator, clinical outcomes considered, sources of key effectiveness and safety parameters, the perspective, analytic horizon, the base-case assumption about treatment cost, and source of funding for the CEA.
b) Point estimates and the proportion of simulations that were cost-effective at a specified threshold.
c) Results of key sensitivity analyses. Because the cost of the intervention may vary by health system or insurer, or over time, a 1-way sensitivity analysis examining how the cost-effectiveness of the intervention varies over a reasonable range of intervention costs can be very helpful for decision-making. Therefore, the supporting text should provide additional information about how sensitive the economic value assessment is expected to be to variation in intervention cost. Inclusion of a figure similar to Figure 4 is encouraged if available. This information

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Table 5. Proposed Structure of the Recommendation Component of Economic Value Statements in ACC/AHA Guidelines

In the examples below, p indicates the current US cost of the intervention; i is the ICER of the intervention relative to the comparator; s is the proportion of probabilistic simulations in which the intervention was costeffective relative to the comparator; t is the cost at which the intervention would be cost-effective relative to the comparator at a cost-effectiveness threshold of $120 000 per QALY gained.
Format 1: If the cost of the intervention is available: "What is the cost-effectiveness of the intervention at its current cost?" In (population/indication), the use of (intervention) at a cost of $*p* (or at current US price or at range of current US prices) is projected to be a/an (class of recommendation) strategy compared with (comparator). Or, when additional quantitative information is available: In (population/indication), the use of (intervention) at a cost of $*p* is projected to be a/an (class of recommendation) strategy compared with (comparator), with an ICER of $*i* per QALY gained (<$120 000 per QALY gained in *s*% of probabilistic simulations).
Format 2: If the cost of the intervention is not available: "What would the cost of the intervention have to be in order for the intervention to meet the cost-effectiveness threshold?" In (population/indication), the use of (intervention) is projected to be a cost-effective strategy compared with (comparator) at a threshold of $120 000 per QALY gained if the cost of the (intervention) is less than $*t*.
Format 3: An alternative format that combines key elements of the 2 formats above into a single recommendation: "What is the cost-effectiveness of the intervention at the current cost and what would the cost of the intervention have to be in order for the intervention to meet the cost-effectiveness threshold?" In (population/indication), the use of (intervention) at a cost of $*p* (or at current US price or at range of current US prices) is projected to be a/an (class of recommendation) strategy compared with (comparator), with an ICER of $*i* per QALY gained (<120 000 per QALY gained in *s*% of probabilistic simulations). The intervention is projected to be cost-effective at a threshold of $120 000 per QALY gained if the cost of the (intervention) is less than $*t*.

ICER indicates incremental cost-effectiveness ratio; and QALY, quality-adjusted life year.

is going to be increasingly important in coming years as the cost of some cardiovascular therapies may change if selected for price negotiation under the Inflation Reduction Act of 2022.[93] Other factors influencing the estimated value of treatment may also be described; for example, whether model results were sensitive to treatment effect heterogeneity,[94] variation in cost effectiveness between subgroups,[95] or assumptions regarding extrapolation of treatment effectiveness beyond trial follow-up.[92] Describing the key inputs that are driving uncertainty in the study estimates of incremental cost-effectiveness should be considered as it may guide future research.

d) Qualitative or quantitative considerations related to the impact of adopting the intervention on population health equity.

e) Budget impact. The supporting text may also include information about the effect of adopting the intervention on total health care spending (see Section 4.2.11, "Budget Impact").

Table 6. Mock-Up of Economic Value Statements

A. Economic Value Statements (adapted from existing ACC/AHA guidelines, with additional information obtained from key publications)		
COR	**LOE**	**Recommendation**
1	A	In patients with symptomatic chronic heart failure with reduced ejection fraction, SGLT2 inhibitors are recommended to reduce hospitalization for heart failure and cardiovascular mortality, irrespective of the presence of type 2 diabetes.[88]
Economic Value: Cost-Effective (High Level of Certainty)		In patients with heart failure with reduced left ventricular ejection fraction, addition of an SGLT2 inhibitor to guideline-directed medical therapy at 2024 prices is projected to be cost-effective compared with prior guideline-directed therapy. OR In patients with heart failure with reduced left ventricular ejection fraction, addition of an SGLT2 inhibitor to guideline-directed medical therapy at a cost of $4192 per patient per year is projected to be cost-effective compared with prior guideline-directed therapy, with an ICER of $68 300 per QALY gained (<$120 000 per QALY gained in 97% of probabilistic simulations).[89]
1	A	In patients with chronic coronary disease, high-intensity statin therapy is recommended with the aim of achieving a ≥50% reduction in low-density lipoprotein cholesterol levels to reduce the risk of major adverse cardiac events.[90]
Economic Value: Cost-Saving (High Level of Certainty)		In patients with chronic coronary disease, the use of generic formulations of maximally tolerated statin therapy is projected to be cost-saving compared with not using statin therapy.[91]
B. Mock-Up Economic Value Statements (not based on current ACC/AHA guidelines)		
Economic Value: Not Cost-Effective (High Level of Certainty)		In patients with transthyretin amyloid cardiomyopathy, treatment with tafamidis at an annual price of $225 000 per person per year is projected to be not cost-effective compared with placebo, with an ICER of $880 000 per QALY gained (<$120 000 per QALY gained in 0% of probabilistic simulations). The use of tafamidis in transthyretin amyloid cardiomyopathy would be cost-effective compared with placebo at a threshold of $120 000 per QALY gained if the cost of tafamidis is <$20 700 per year.[92]
Economic Value: Indeterminate (Insufficient Evidence)		In patients with a body mass index ≥27 kg/m^2 and established atherosclerotic cardiovascular disease but without diabetes, the cost effectiveness of semaglutide for secondary prevention of cardiovascular disease is uncertain.

ICER indicates incremental cost-effectiveness ratio; and QALY, quality-adjusted life year.

f) The supporting text may reference high-quality CEAs from other countries, particularly if the comparison with US-based analyses generates important insights regarding the value of the intervention being evaluated.

7. EXPERTISE IN COST-EFFECTIVENESS RESEARCH NEEDED FOR GUIDELINES

As described above, the role of the guideline committee is to review the published cost-effectiveness literature,

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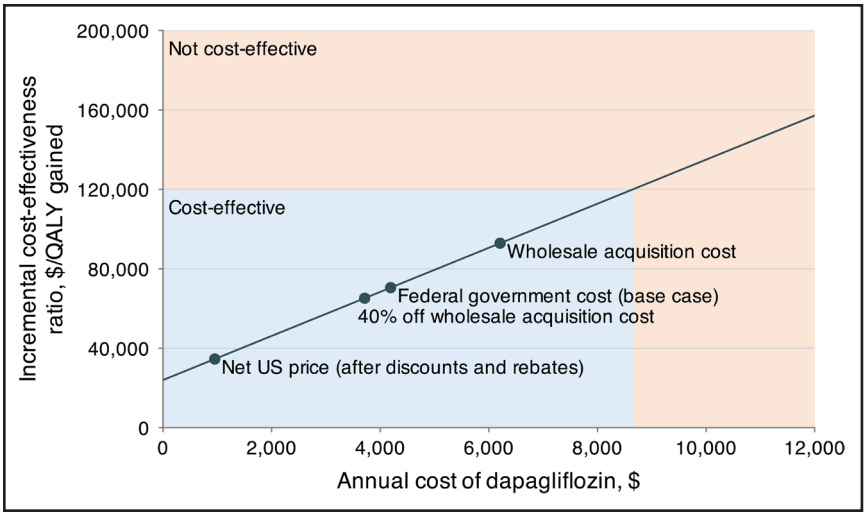

Figure 4. One-Way Sensitivity Analysis on the Price of the Intervention.
In this example, the ICER of adding dapagliflozin to prior guideline-directed medical therapy for heart failure with reduced ejection fraction varies with the cost of dapagliflozin. ICER indicates incremental cost-effectiveness ratio; and QALY, quality-adjusted life year. Modified with permission from Isaza et al.[89] Original material distributed under a CC-BY license (https://creativecommons.org/licenses/by/4.0/).

determine the quality of each study, and summarize data across evaluations if there are multiple studies. Thus, each guideline committee will need at least 1 member with expertise in reviewing published economic studies. The selection process for the economic expert(s) on a writing committee should consider training, experience, and scholarship in economic evaluations of cardiovascular interventions. The number of individuals with economic expertise required to serve on the committee will depend on the number of clinical areas the guideline wishes to evaluate. For most guidelines, a single expert may suffice. However, the inclusion of 2 or more experts may be preferrable for guidelines requiring the evaluation and synthesis of a large number of CEAs (and may facilitate assessment of evidence quality). Conflicts of interest must be carefully managed in any assessment of economic value. While all relevant conflicts should be disclosed and reviewed, not all relationships with industry or payers should be considered disqualifying. For instance, contracts through academic institutions should not be regarded the same as significant equity interests or other personal or family financial interests, which would be disqualifying. This may also influence the number of committee members needed to evaluate the economic value of all desired care areas.

8. KEY CHALLENGES AND SPECIAL CONSIDERATIONS

8.1. Rare Disease Drugs and Gene Therapies

Recent scientific advances have led to the approval of first-in-class treatments for several rare cardiovascular diseases (defined in the United States as affecting fewer than 200 000 individuals). The high list prices of these therapies, such as tafamidis for transthyretin amyloidosis ($250 000 per year), invariably result in unfavorable cost-effectiveness determinations when using traditional $100 000 to $150 000/QALY thresholds.[92] Some bodies have proposed the use of broader ranges for cost-effectiveness thresholds for rare diseases, as is done in some countries,[96,97] while others have noted that US policy already provides substantial financial incentives for rare disease therapies, which should be counterbalanced with value-based pricing to ensure access to newly approved therapies.[98] Similar arguments have been made for one-and-done gene therapies, where a single upfront treatment may produce a durable benefit and avert the need for lifelong treatment with conventional pharmacotherapy.

The writing committee recognizes the trade-off between the need to incentivize research and development of novel therapies that will help patients in the future while also ensuring that currently approved treatments are accessible to all patients who would benefit from them. Therefore, we propose that rare disease drugs and gene therapies be evaluated using the standard cost-effectiveness threshold ($120 000 per QALY gained) for the base-case scenario but with heightened consideration of contextual factors around severity of disease, unmet need, health equity, and access to care. Economic evaluations of gene therapies should report the results of sensitivity analyses that vary the durability of effectiveness and incorporate delayed onset of adverse events. The supporting text accompanying economic value statements for rare disease and gene therapy interventions may be used to comment on the innovation-access trade-off. Where available, results of distributional CEAs, which formally incorporate equity considerations into the analysis, may provide additional insight into the cost effectiveness of treatments for rare diseases.[99,100]

8.2. Medical Devices

Evaluating the cost-effectiveness of medical devices can pose unique challenges. Because their regulatory requirements differ from those of pharmaceuticals, new medical

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devices may enter the market and gain widespread adoption without having been evaluated in adequately powered randomized trials.[101] Clinical outcomes may vary based on the expertise of the clinician using/implanting the device, unique patient characteristics (eg, device-specific anatomic factors), and the experience of health system. Moreover, small modifications of the device seldom trigger a formal reevaluation, yet, over time, the device in clinical use may be substantially different from the version originally tested in clinical trials (eg, implantable defibrillators or transcatheter aortic valves).[102] High-quality CEAs should be transparent about the source of effectiveness and safety parameters (and, if substantial changes have occurred over time, the version of the device that these inputs refer to). They should also evaluate how sensitive their conclusions are to uncertainty in these parameters. Device prices can vary widely across health systems based on the discounts they are able to negotiate (which are determined by the purchasing power of the health system and the presence of competition in the market). CEAs should use the price net of discounts where available or the list prices when the net price is not available. In either case, device costs should be varied widely in 1-way sensitivity analyses to evaluate the effect on the ICER.[103] In situations where device costs are truly opaque, the economic value statement should report the threshold prices at which the device would be expected to be cost-effective (Format 2 shown in Table 5).

8.3. Policy Interventions

Where relevant, guidelines may report the findings of CEAs that evaluate the cost-effectiveness of health policy interventions (eg, implementation of a tax on sugar-sweetened beverages).[104,105] As with CEAs evaluating clinical interventions, those evaluating policy interventions should use an analytic horizon that is sufficiently long to fully capture the outcomes of the policy while balancing legitimate concerns about uncertainty in long-term population health outcomes. Careful attention must be paid to the source of key inputs because policies are seldom evaluated in randomized trials. For most policy interventions, estimates of short-term effectiveness and any unintended adverse consequences are estimated from observational analyses, including quasi-experimental study designs. Moreover, CEAs of policy interventions rely on modeling assumptions to forecast long-term health and economic outcomes.[106] Because of the numerous sources of uncertainty in a CEA of this nature, careful expert review of the underlying assumptions and extensive sensitivity analyses to evaluate the impact of uncertainty on study findings is necessary.

8.4. Interpreting Older CEAs

When relying on results from older CEAs, ICERs should be updated prior to interpretation. Interval developments may affect the cost-effectiveness of an intervention over time, including updated estimates of the effectiveness and safety of the intervention or the comparator, changes in clinical practice, or the emergence of new alternatives. The costs of drugs or devices can also change substantially over time (eg, marked declines in drug prices when generic formulations become available), as can overall health care costs (eg, large increases in the cost of treating acute cardiovascular events over the past 2 decades). Finally, the baseline risk in the population may change over time, which can alter the absolute benefit derived from an intervention. For instance, the advent of effective medical therapy may reduce the risk of arrhythmic death, likely lowering the absolute benefit of an implantable defibrillator for primary prevention. Collectively, these changes can markedly alter the economic value of an intervention.[107]

Ideally, older CEAs should be updated with new values of key input parameters, but updated analyses are not always available at the time when guidelines are being prepared. Adjusting the cost of the intervention and comparator(s) to contemporary US costs and inflating components of cost to a contemporary year using the indicators described in Section 4.2.6, "Costs," may be a reasonable starting point, provided there have been no major interval changes in the estimates of safety and effectiveness of the intervention or the comparator.[108] If relying on this approach, analysts should inflate the cost of the intervention and the comparator separately from other health care costs, as these may have had different trajectories over the period since the original analysis was published. The original results should be reported together with the updated value version so readers can see and verify the results of the calculations.

In some cases, it may be reasonable to rely on the results of previously reported sensitivity analyses in order to account for any interval changes in the real (ie, inflation-adjusted) cost of the intervention. However, this approach is unlikely to work for all CEAs; in some cases, a systematic update (ie, re-running the analysis with updated inputs) may be the only acceptable quantitative solution.

Given that the writing committee is not likely to have the resources to perform additional analyses, a qualitative assessment of interval changes—and whether they would be expected to increase or decrease the ICER—may be reasonable. Older CEAs can still generate useful insights to inform economic value statements, but their findings merit cautious interpretation. The supporting text accompanying the economic value statement should provide the reader with the process used to quantitatively or qualitatively update older CEAs on which economic value statements are based.

8.5. Patient Out-of-Pocket Costs

Economic value statements have historically focused on the health care sector perspective (ie, they consider

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all health care–related costs, regardless of who pays for them). As costs for cardiovascular care are on the rise, some of these costs are passed on to patients as out-of-pocket costs. More than 29% of insured adults and more than 60% of uninsured adults have delayed or forgone health care due to rising medical costs.[109] Among Medicare beneficiaries with cardiovascular disease or cardiovascular risk factors, approximately 1 million adults have annual out-of-pocket costs in excess of $2000.[110] This is likely an underestimate, as many patients with high out-of-pocket costs choose to forego medications they need. In a meta-synthesis, clinicians also believed that inadequate insurance contributed to inequitable clinical decision-making for minoritized groups,[111] as uninsured rates remain substantially higher among American Indian, Alaska Native, Hispanic, Native Hawaiian, other Pacific Islander, and Black populations than in White populations.[112] Implementation of the Inflation Reduction Act of 2022 may help by expanding access to Medicare's Low-Income Subsidy (starting in 2024), capping out-of-pocket costs at $2000 per person (starting in 2025) and allowing costs to be spread over the year rather than requiring a single lump payment.[113] In addition, 7 of the first 10 drugs selected for price negotiation are for the treatment of cardiovascular disease or cardiovascular risk factors.[93] While these reforms will lower cost-sharing for Medicare Part D beneficiaries and costs to Medicare, they are unlikely to benefit patients with other forms of insurance.

We suggest that guideline writing committees consider the inclusion of 1 or more recommendations regarding the need for clinicians to discuss out-of-pocket costs with patients at the time of initiation of a new therapy and at regular intervals thereafter, with the goal of identifying financial strain resulting from out-of-pocket costs and preempting cost-related nonadherence.

9. SUMMARY

A summary of the methodological guidance from this writing committee is presented in Table 7.

10. FUTURE DIRECTIONS

10.1. Reference Case Inputs

Comparability across CEAs could be enhanced by the use of a common set of transition probabilities, costs, and utility parameters in the reference case. Ideally, these parameters would be derived from publicly available datasets by investigators who also publish their statistical code so that they may be replicated and adapted by other researchers.[114,115] Such a core of common input parameters would increase comparability across model-based economic evaluations as it would minimize differences in results across different studies resulting from differences in underlying inputs. Funding agencies should consider investing in a core of input parameters for cardiovascular CEAs, as the Cancer Intervention and Surveillance Modeling Network funded by the National Cancer Institute has done for oncology CEAs.[116,117]

10.2. Heterogeneity in CEA

Economic value statements in guidelines typically rely on the average cost-effectiveness of the intervention across the population, equivalent to the average treatment effect estimated in a clinical trial. However, just as heterogeneity in the treatment effect (on the absolute scale) is almost always present across subgroups of the study population, there may also be meaningful heterogeneity in cost-effectiveness.[118–121] Some subgroups may derive a larger or smaller absolute benefit than suggested by the average treatment effect, and some subgroups may have a higher burden of adverse events than the overall treatment-eligible population. This may be driven by differences in the population characteristics, cardiovascular risk in the untreated population, effectiveness or safety of the intervention, implementation of the intervention, or competing risks. Three instances arise where CEA heterogeneity becomes relevant. First, policymakers may be value-conscious in some jurisdictions and consider heterogeneity in cost-effectiveness to target high-benefit populations.[118,121] Second, benefit design may include low or no out-of-pocket costs and expanded access in settings where a therapy is cost-saving or cost-effective.[119,121] Third, heterogeneity assessment in CEAs can identify knowledge gaps (eg, about implementation in certain populations) and help prioritize future research.[119,122] The supporting text accompanying an economic value statement may include information about heterogeneity in cost-effectiveness across key subgroups, particularly if this information may guide strategies to advance health equity. At the same time, limitations of subgroup analyses (eg, limited precision in effectiveness estimates) should be acknowledged where relevant.[123]

10.3. Other Approaches to Evaluating Cost Effectiveness

The use of a cost-effectiveness threshold is 1 way to evaluate the economic efficiency of a new intervention and guide its adoption. At least 2 other alternative approaches may be considered. One is to identify a benchmark intervention, ie, an intervention that a society explicitly or implicitly has considered worthy of investment.[124] By extension, any intervention that has an ICER that is equal to or less than the ICER of the benchmark intervention would also be considered economically attractive. For many years, mandatory funding for hemodialysis codified in the US Medicare program was used as an example of a benchmark (even though Medicare's coverage of hemodialysis was not based on its economic value). In this approach, each

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Table 7. Summary of Methodological Guidance Included in the 2025 ACC/AHA Cost-Value Methodology Statement

Expertise	At least 1 member with expertise in reviewing and, ideally, performing economic studies related to cardiovascular disease should serve on each guideline committee. The number of individuals will depend on the number of clinical areas the guideline committee wishes to evaluate. Conflicts of interest must be carefully managed.
Systematic search	A careful review of the literature and consultation with subject matter experts in the field is necessary to identify all relevant economic evaluations. Both trial-based economic evaluations and population-based economic evaluations can be used to inform economic value statements. Economic value statements should be based on CEAs that adopted a health care sector perspective, and results from the societal perspective, if available, may be reported in the supporting text. Economic value statements should rely exclusively on studies that use costs that are observed or estimated in the US population (costs measured in other populations and converted to US dollars are not usable for this purpose). Economic value statements should be based on CEAs that measure health outcomes in terms of QALYs gained or life years gained, and, where available, equal value of life years, and health years in total.
Quality of evidence	Established quality assessment tools, such as the CHEQUE, can help evaluate the methodological rigor of a health economic study. Economic value statements should rely on high-quality studies. The reference case described in this statement may help identify relevant results from a high-quality cardiovascular CEA (Table 2). When multiple CEAs are available on a given topic, the supporting text should clarify the underlying rationale for why some studies were considered more relevant than others.
Intervention	Guidelines should consider including economic value statements for: Interventions that receive Class 1 and 2A recommendations in the clinical guidelines Interventions with potential for substantial impact on total health care spending Clinical guidelines should include economic value statements about all interventions for which economic evaluations of adequate quality are available, including preventive, diagnostic, and therapeutic strategies, and approaches to health care delivery relevant to the clinical guidelines. The decision about whether or not an economic value statement is appropriate should rely on the judgment of clinical and economic experts on the guideline writing committee (based on the clinical and population health relevance, cost, available evidence, etc.)
Reference case	Economic value statements should emphasize high-quality CEAs that adhere to the specifications of the reference case in this statement. Results should be reported as ICER per QALY gained. Where available, net monetary benefit and net health benefit metrics may be reported as a function of the cost-effectiveness threshold.
Economic value statement	Guidelines should use the "Economic Value Statement" instead of the previously used term "Value Statement" to clarify that cost effectiveness is only 1 component of value. Because the economic value statement is meant to be interpreted in the context of the clinical recommendation, the economic value statement should immediately follow the clinical recommendation for the intervention. Each economic value statement should consist of the class of recommendation, level of certainty, and recommendation (Tables 3–6). Interventions are considered to be cost-effective at a threshold of $120 000 per QALY gained. The recommendation should include the population, intervention, comparator, cost of the intervention, and ICER (with % of simulations below the accepted threshold, if available) (Table 5). When the economic value statement must rely on studies that examine a population that is systematically different from the target population, this should be reflected in the wording of the economic value statement.
Sensitivity analyses	Key findings from deterministic and probabilistic sensitivity analyses should be reported in the supporting text.
Supporting Text	Supporting text should summarize the high-quality CEAs that were used to generate the economic value statement, including the study population (and whether it differs from the corresponding population for the guideline statement), intervention, comparator, clinical outcomes considered, sources of key effectiveness and safety parameters, perspective, analytic horizon, base-case assumption about treatment cost, and source(s) of funding for the CEA. Point estimates of the ICER and the proportion of simulations that were cost-effective at a specified threshold should be reported. Insights from key sensitivity analyses should be reported (eg, key inputs that drove uncertainty in the study estimates). A figure of how the ICER varies with the cost of the intervention may be included if available. It may be reasonable to highlight subgroups in whom the ICER is substantially higher versus lower than in the overall population (see Section 10.2). Considerations related to health equity not already reflected in the economic value statement should be discussed in the supporting text. Information about the effect of adopting the intervention on total health care spending (ie, the budget impact) should be included if available. The supporting text may reference high-quality CEAs from other countries, particularly if the comparison with US-based analyses generates important insights regarding the value of the intervention being evaluated.
Miscellaneous	Rare disease drugs and gene therapies should be evaluated using the standard cost-effectiveness threshold ($120 000 per QALY gained) but with heightened consideration of contextual factors around health equity and access to care. One or more recommendations should highlight the need for clinicians to discuss out-of-pocket costs with patients at the time of initiation of a new therapy and at regular intervals thereafter, with the goal of identifying financial strain resulting from out-of-pocket costs and preempting cost-related nonadherence.

CEA indicates cost-effectiveness analysis; CHEQUE, Criteria for Health Economic Quality Evaluation; ICER, incremental cost-effectiveness ratio; and QALY, quality-adjusted life year.

intervention is evaluated relative to its own comparator and the resulting ICER is compared with the benchmark. While benchmarking is an attractive approach in theory, investment decisions often involve considerations beyond cost-effectiveness (such as equity implications or political factors), so identifying a single benchmark intervention for

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a health system may be challenging. Another approach involves evaluating all possible interventions and listing them in increasing order of ICERs, creating a "league table."[72,125,126] The health system starts by investing in the most cost-effective intervention and moves down the list until the health care budget is exhausted. The CEA registry maintained by Tufts Medical Center is a database of more than 10 000 CEAs that allows the comparison of ICERs across a range of interventions, diseases, and conditions and is an example of a generic league table.[127] Experts have proposed the league table approach for use in health systems with a fixed budget, but this approach is impractical for the fragmented US health system. However, individual payers or programs may operationalize a more limited form of this approach to optimize resource allocation.

10.4. Coupling Performance Measures With Economic Value

The current approach to guideline development is that the class of recommendation for the clinical recommendation is based on the strength of the clinical evidence and does not consider the economic value of the intervention under consideration (or other factors such as absolute risk reduction and the expected population impact of the intervention). In contrast, the economic value statement explicitly considers incremental cost-effectiveness, which is a composite of both health and economic outcomes. We propose that guideline writing committees continue this practice for now. However, it is plausible that in the future, economic value assessments may inform class of recommendation of the clinical recommendation, with the Class 1 designation reserved for interventions that have strong clinical evidence of health benefit and are also cost-effective at the accepted threshold. Alternatively, when procedures or interventions receive a Class 2 recommendation (ie, probably recommended and may be appropriate for some patients), the economic value assessment may help determine whether an intervention receives a Class 2A designation (moderate benefit and meets the cost-effectiveness threshold) versus a Class 2B designation (benefit only marginally exceeds risk or does not meet the cost-effectiveness threshold). This approach would provide an incentive for stakeholders to price interventions according to the value they generate for patients and society.

11. CONCLUSION

A CEA quantifies the efficiency of a health intervention by explicitly examining the resources consumed relative to the health benefits produced by the adoption of the intervention compared with the alternative strategy. Therefore, the results of high-quality CEAs, coupled with information about the feasibility of the intervention and its effect on health equity, can help health system and policy leaders make informed decisions regarding optimal allocation of finite societal resources. The 2025 Cost/Value methodology statement provides guidance for synthesizing high-quality evidence regarding the cost-effectiveness of cardiovascular interventions and developing economic value statements for clinical guidelines. It proposes a cost-effectiveness threshold of $120 000 per QALY for evaluating the cost-effectiveness studies of health interventions in the United States and offers a new framework for communicating the level of certainty in economic value statements. It highlights special considerations as they relate to cardiovascular drugs and devices, emphasizes health equity considerations when interpreting CEA findings, and defines a reference case for future CEAs of cardiovascular interventions.

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ARTICLE INFORMATION

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Appendix 1. Author and Peer Reviewer Relationships With Industry and Other Entities (Comprehensive)—2025 ACC/AHA Statement on Cost/Value Methodology in Clinical Practice Guidelines (Update From 2014 ACC/AHA Statement)

Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Dhruv S. Kazi (Chair)	Beth Israel Deaconess Medical Center—Associate Professor of Medicine	• PCORI	None	None	• AHA • AHRQ* • Institute for Clinical and Economic Review* • NIH/NHLBI	• AHA† • Boston Scientific Corporation* • NIH*	None
Abdul R. Abdullah‡	AHA/ACC—National Director, Science & Medicine, Guideline Science	None	None	None	None	• AHA/ACC salaried employee*	None
Suzanne V. Arnold	Saint Luke's Mid America Heart Institute—Cardiologist	None	None	None	• FDA* • NHLBI*	• *JAMA Network Open J*	None
Anirban Basu	University of Washington School of Pharmacy—Director & Professor of Health	• Salutis Consulting*	None	None	None	None	None
Brandon K. Bellows	Columbia University—Assistant Professor of Medical Sciences	None	None	None	None	• NHLBI (PI)* • School of Medicine, University of Alabama at Birmingham*	None
Khadijah Breathett	Indiana University School of Medicine—Associate Professor of Medicine	None	None	None	• HRSA* • Indiana University* • NIH/NHLBI**	• *American Heart Journal*† • Black Health Matters† • *Circulation: Cardiovascular Quality and Outcomes** • Indiana University* • NHLBI*	None
Derek S. Chew	University of Calgary—Assistant Professor; Alberta Health Services—Cardiac Electrophysiologist	None	None	None	• Canadian Institutes of Health Research* • Heart and Stroke Foundation of Canada*	None	None
David J. Cohen	Cardiovascular Research Foundation—Director of Clinical and Outcomes Research; St. Francis Hospital & Heart Center—Director of Academic Affairs	• Abbott Vascular* • Boston Scientific* • Edwards Lifesciences* • Medtronic*	None	None	None	• Abbott Vascular* • Boston Scientific* • Cardiovascular Systems • CathWorks, Inc.* • Corvia* • Edwards Lifesciences* • iRhythm Technologies* • JC Medical* • JenaValve* • Medtronic* • Philips*	None
Colette DeJong	Stanford University School of Medicine—Clinical Instructor; VA Palo Alto Health Care System—Staff Cardiologist; iRhythm Technologies (spouse)	None	None	• iRhythm Technologies*	• Center for AIDS Prevention Studies, UCSF* • Merck* • VA Palo Alto Health Care System*	• iRhythm Technologies*	None

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Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Paul A. Heidenreich	Stanford University–Professor and Vice Chair for Quality, Department of Medicine; VA Palo Alto Health Care System–Chief of Medicine	None	None	None	• Department of Veterans Affairs* • Gordon Betty Moore Foundation*	• ICER • VA Palo Alto Health Care System	None
Susan Hennessy	UCSF–Research Data Analyst, Epidemiology & Biostatistics	None	None	None	• Wyeth (SYRMOUNT)*	• Novartis (Global Heart Hub)†	None
Inmaculada Hernandez	University of California, San Diego, Skaggs School of Pharmacy and Pharmaceutical Sciences–Professor, Division of Clinical Pharmacy	• Bristol Myers Squibb • Pfizer	None	None	None	• Curta*	None
Nicolas Isaza	Beth Israel Deaconess Medical Center–Fellow, Cardiovascular Medicine	None	None	None	None	None	None
Karen E. Joynt Maddox	Washington University School of Medicine in St. Louis–Associate Professor of Medicine	• Centene Corporation*	None	None	None	• *Journal of the American Medical Association** • NHLBI* • NIA* • NIH*	None
Ann Marie Navar	UT Southwestern Medicine–Associate Professor, Internal Medicine	• Amgen* • Arrowhead Pharmaceuticals • AstraZeneca† • Bayer* • Boehringer Ingelheim† • Bristol Myers Squibb† • Eli Lilly and Company* • Esperion Therapeutics* • Genentech* • Idorsia (spouse)* • Janssen Pharmaceuticals* • Johnson & Johnson* • Merck* • Miga Health • NewAmsterdam Pharmaceuticals* • Novartis† • Novo Nordisk* • Pfizer • Sanofi† • Silence Pharmaceuticals	None	• Miga Health	• Amgen* • Amgen (Co-PI, spouse) • Arrowhead Pharmaceuticals (DSMB) • Boston Scientific (DSMB, spouse)* • Eli Lilly* • Esperion Therapeutics* • Esperion Therapeutics (Co-PI, spouse)* • Johnson & Johnson (spouse)* • Merck* • NewAmsterdam Pharmaceuticals* • Novartis • Novo Nordisk* • Silence Therapeutics	• American Medical Association • American Society for Preventive Cardiology	None
Ankur Pandya	Harvard T.H. Chan School of Public Health–Associate Professor of Health Decision Science	None	None	• Roche (spouse)	None	None	None

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Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Kendra D. Sims	Boston University School of Public Health–Postdoctoral Associate, Epidemiology; UCSF–Postdoctoral Scholar, Epidemiology & Biostatistics	None	None	None	None	• NIH*	None
Merilyn S. Varghese	Yale University School of Medicine–Assistant Professor; VA Connecticut Healthcare System–Cardiologist	None	None	None	• US Department of Veterans Affairs*	• AHA*	None
Liesl Zühlke	University of Cape Town–Professor, Division of Pediatric Cardiology; South African Medical Research Council–Vice President Extramural Research & Internal Portfolio	None	None	None	None	None	None
Peer Reviewers							
Daniel Mark	Duke University School of Medicine–Professor of Medicine; Duke Clinical Research Institute–Member	• Boehringer Ingelheim* • CeleCor • Novartis	None	None	• AHA* • HeartFlow* • Merck* • NHLBI* • Novo Nordisk*	• Elsevier	None
Jason Wasfy	Harvard Medical School–Associate Professor; Massachusetts General Hospital–Associate Physician	• Institute for Clinical and Economic Review • *Journal of the American College of Cardiology* • Point32Health	None	None	• National Football League Players Association (spouse)* • NIH*	• *Circulation: Cardiovascular Quality and Outcomes** • Lancet Commission on Ischemic Heart Disease†	• Defendant, cardiovascular care, 2024*

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†No financial benefit.

‡Abdul R. Abdullah is an AHA/ACC joint staff member and acts as the Director for the 2025 ACC/AHA Cost/Value Methodology Statement. No relevant relationships to report. Nonvoting author on measures and not included/counted in the RWI balance for this committee.

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